

DANISH ILYAS

AI Agent & Python Developer | Automation Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

AI & Python Developer with hands-on experience shipping production systems for real clients: AI agents, business-automation bots, API integrations and full-stack web applications. Built a bilingual WhatsApp Cloud API bot that runs a live ERP/POS from chat, integrated a retail POS with the FBR (AJK-IRD) government tax API, and achieved 99.8% accuracy on a multimodal medical-diagnosis model. Comfortable across two ecosystems — Python (FastAPI, Django, LangChain, PyTorch) and C#/.NET (ASP.NET Core, SQL Server, Entity Framework). Currently pursuing BS Computer Science at UCP while delivering paid client work as a freelancer.

SKILLS & TECHNOLOGIES

AI / LLMs: OpenAI GPT-4, LangChain, RAG Pipelines, Prompt Engineering, AI Agents, LLM Integration

ML / Deep Learning: PyTorch, TensorFlow, Scikit-learn, ResNet50, BERT / Bio_ClinicalBERT, OpenCV, Pandas, NumPy

Backend: FastAPI, Django, Flask, ASP.NET Core MVC, REST APIs, WebSockets, JWT Auth, Celery, ARQ, Redis

Automation & APIs: WhatsApp Cloud API, Meta Graph API, TikTok API, FBR / AJK-IRD POS API, Webhooks, Selenium, yt-dlp + FFmpeg, Windows Services, Scheduled Jobs

Databases: PostgreSQL, SQL Server, SQLite, SQLAlchemy, Entity Framework Core, Alembic, Redis

Frontend: HTML5, CSS3, JavaScript, HTMX, Tailwind, Jinja2, Razor / Bootstrap, React (basics), Streamlit

DevOps / Tools: Docker, Caddy, Git & GitHub, Visual Studio, VS Code, Jupyter, Postman

Languages: Python (Advanced), C# (Intermediate), C++ (Intermediate), SQL, JavaScript

PROFESSIONAL EXPERIENCE

Freelance Software Engineer — AI, Automation & Web

Fiverr ([asad_ilyas1](#)) & Direct Clients | Remote | 2024 – Present

- Maintain and extend a live ASP.NET Core ERP/POS suite for agri-trading, textile and retail clients — sales, purchases, inventory, accounting, ledgers and reporting modules on SQL Server.
- Built a **bilingual (English/Urdu) WhatsApp business bot** on top of that ERP so owners and customers can place orders, check balances and pull daily reports directly from chat — removing the need for desktop access.
- Integrated the POS with the **FBR (AJK-IRD) government tax invoicing API**, including buyer NTN/CNIC validation, HS-code (PCT) mapping and tax-exempt handling for legally compliant invoices.
- Currently architecting a **multi-platform content syndication system** that mirrors YouTube uploads to Facebook Pages and TikTok automatically for a content agency.
- Maintain a **4.9★ Fiverr rating across 44+ verified reviews** with 100% on-time delivery.

FEATURED PROJECTS

Content Syndication Automation Platform — Client Project (In Development)

Designed and building an end-to-end system that detects YouTube creators missing from Facebook/TikTok and auto-mirrors every new upload to their pages with identical title, description, tags and thumbnail. Includes per-creator Manual / System / Paused migration modes, CSV bulk import, background download-and-publish workers, and a planned unified insights dashboard for cross-platform performance.

Stack: Python 3.12 · FastAPI · HTMX · Jinja2 · PostgreSQL · Redis + ARQ · yt-dlp + FFmpeg · Meta Graph API · TikTok API · Docker · Caddy

WhatsApp Business Bot for ERP / POS — Client Project (Live)

Conversational bot on the WhatsApp Cloud API that turns the ERP into a chat interface. Separate customer and admin menus: place orders, price list, order history, balance and ledger for customers; daily sales, Z-report, pending orders, stock alerts, customer ledgers, add-product and record-purchase flows for admins. Features a stateful multi-step conversation engine, webhook message handling, full English/Urdu localisation with per-number language memory, PDF invoice/report generation uploaded as WhatsApp media, and a scheduled notification service.

Stack: C# · ASP.NET Core · WhatsApp Cloud API · Webhooks · Entity Framework Core · SQL Server · QuestPDF

FBR (AJK-IRD) Tax Integration for Retail POS — Client Project (Live)

Integrated a retail POS with Pakistan's FBR AJK-IRD digital invoicing API so every sale is reported to the tax authority in real time. Implemented customer NTN/CNIC/STRN management with searchable lookup at the sale window, walk-in customer fallback, tax-exempt product flags, HS/PCT code mapping, and a bulk Excel import/export tool for product and invoice data.

Stack: C# · ASP.NET Core MVC · FBR/AJK-IRD REST API · Entity Framework Core · SQL Server · EPPlus

Multimodal Medical Diagnosis AI — 99.8% Accuracy

Late-fusion deep learning system combining chest X-ray images with clinical notes to classify 14 lung conditions. Fused a ResNet50 image encoder with a Bio_ClinicalBERT text encoder and served predictions through a FastAPI backend with a Streamlit diagnosis interface.

Stack: *PyTorch · FastAPI · Streamlit · ResNet50 · Bio_ClinicalBERT*

Hospital Management System — Production Ready

Full-stack HMS covering 20+ database tables and 5 role-based access levels, with JWT authentication, real-time WebSocket notifications, SMS alerts, appointment and billing workflows, and a fully documented REST API.

Stack: *FastAPI · PostgreSQL · SQLAlchemy · Alembic · React · WebSocket · JWT*

Job Agent — AI Resume Matcher & Career Tool

LangChain-powered agent that scrapes live job postings, scores resume-to-job fit, generates tailored cover letters and produces skill-gap analysis reports. Fully dockerised for deployment.

Stack: *FastAPI · LangChain · OpenAI · Docker · PDF Parsing*

EduAssist — AI-Powered Learning Assistant

Educational chatbot built on GPT-4 with automated test evaluation, OCR-based grading of photographed answer sheets, adaptive learning paths and a student performance analytics dashboard.

Stack: *FastAPI · OpenAI GPT-4 · Streamlit · OCR · SQLite*

E-Commerce Price Tracker with Automated Alerts

Tracks Amazon and Daraz product prices every 6 hours through Celery Beat and emails the user when a target price is hit. Includes price-history charts, anti-bot scraping with Selenium and a CLI management interface.

Stack: *FastAPI · PostgreSQL · Celery · Redis · Selenium · SMTP · Chart.js*

Smart Parking Management System — University Capstone

Parking system with a self-contained C++ HTTP server backend (no framework) exposing a JSON REST API for slot allocation, check-in/check-out and fee calculation, paired with a responsive vanilla-JavaScript dashboard.

Stack: *C++ · cpp-httplib · REST API · JavaScript · HTML5 / CSS3*

Restaurant Ordering & Booking Website

Production-ready Django site with online ordering and live cart, table reservations, moderated customer reviews, WhatsApp/Call quick actions, Google Maps integration and a complete admin panel — hardened with CSRF, XSS and input-validation protections.

Stack: *Django · PostgreSQL · JavaScript · AJAX · Bootstrap*

Heart Disease Prediction System

Comparative ML study benchmarking Logistic Regression, Random Forest and XGBoost with feature-importance analysis, evaluation dashboards and deployable .pkl model artefacts.

Stack: *Python · Scikit-learn · XGBoost · Pandas · Matplotlib*

EDUCATION

BS Computer Science (Ongoing)

University of Central Punjab (UCP), Lahore | 2024 – Present

CERTIFICATIONS

- **The AI Engineer Course 2025** — Udemy

KEY HIGHLIGHTS

- Shipped **live client software in production use** — ERP/POS modules, a WhatsApp business bot and a government tax-API integration.
- Delivered **15+ projects** across AI agents, ML models, automation systems, full-stack web apps and desktop/service tooling.
- **99.8% accuracy** on a multimodal medical diagnosis task using late-fusion deep learning.
- Comfortable in **both Python and C#/.NET stacks**, from ML notebooks to enterprise MVC applications on SQL Server.
- Experienced with **third-party API integration at scale** — Meta Graph, WhatsApp Cloud, TikTok, OpenAI and government (FBR) APIs.
- Bilingual product work (English / Urdu) with full localisation and RTL-aware content.